Dataset Processing and Selection Rationale

BOMNI and PIROPO Datasets for Fisheye Pedestrian Detection

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February 16, 2026

1 Introduction

This document describes the unified processing pipeline applied to BOMNI [1] and PIROPO [2] datasets for evaluating projection configurations in fisheye pedestrian detection. Both datasets use rotated bounding box annotations provided by Tamura et al. [3]. Our processing follows a consistent two-step workflow with manual quality verification to ensure evaluation integrity.

2 Common Processing Pipeline

Both BOMNI and PIROPO undergo identical processing steps, ensuring methodological consistency for fair cross-dataset comparison.

2.1 Step 1: Automated Preparation

2.1.1 Annotation Format Conversion

Input: Tamura XML annotations in repurposed Pascal VOC format

Process:

1. Parse XML files to extract rotated bounding box parameters
2. Decode center and dimensions from xmin/ymin/xmax/ymax fields:
   - $center_x = (xmin + xmax)/2$
   - $center_y = (ymin + ymax)/2$
   - $width = xmax - xmin$
   - $height = ymax - ymin$
3. Compute rotation angle: $\theta = \arctan 2(dx, -dy)$ where (dx, dy) is the vector from fisheye center to bbox center
4. Convert to standard JSON format with explicit fields: `center_x`, `center_y`, `width`, `height`, `angle`, `class_name`

Output: JSON annotation files with precomputed rotation angles

2.1.2 Why Tamura's Format Requires Conversion

Tamura's format limitation: The original Pascal VOC XML format does *not* include an explicit rotation angle field. Tamura et al. repurposed the standard Pascal VOC bounding box fields (xmin, ymin, xmax, ymax) to encode the center position and dimensions of rotated bounding boxes, but the rotation angle itself is *omitted*.

Implicit angle encoding: Tamura assumes all pedestrian bounding boxes follow the fisheye radial distortion pattern. The rotation angle can be reconstructed geometrically:

$$\theta = \text{arctan2}(dx, -dy) \quad \text{where} \quad \begin{cases} dx = \text{bbox\_center}_x - \text{fisheye\_center}_x \\ dy = \text{bbox\_center}_y - \text{fisheye\_center}_y \end{cases} \tag{1}$$

This (dx, dy) vector points from the fisheye center to the pedestrian's center, and its angle determines the bbox rotation (height axis aligned radially).

Why this design is problematic:

1. **Computational redundancy**: The rotation angle must be recalculated via `arctan2()` *every single time* annotations are loaded—during training (thousands of epochs × thousands of images), evaluation, and visualization. This represents massive redundant computation for a value that never changes.

2. **Format ambiguity**: Standard Pascal VOC parsers will misinterpret the annotations as axis-aligned boxes, yielding incorrect results unless the user knows about Tamura's repurposing convention.

3. **Lack of flexibility**: The implicit angle encoding enforces the radial constraint. If non-radial rotations are needed (e.g., for data augmentation or correcting annotation errors), the format cannot represent them.

Why Tamura chose this format: Likely reasons include:

- **Pascal VOC compatibility**: Adhering to an established annotation standard (Pascal VOC) for backwards compatibility with existing tools, despite the standard not supporting rotation.

- **Legacy tooling**: Reusing existing annotation tools that output Pascal VOC format rather than developing custom tooling.

- **Enforcing radial constraint**: Ensuring all annotators follow the fisheye radial pattern by making angle calculation automatic rather than manual.

Our conversion rationale: We address these limitations by:

1. **Precomputing angles once**: Calculate $\theta = \text{arctan2}(dx, -dy)$ during the one-time conversion step and store it explicitly in JSON. This eliminates redundant trigonometric operations during training/evaluation (compute once, use thousands of times).

2. **Explicit, self-documenting format**: Our JSON schema (`center_x`, `center_y`, `width`, `height`, `angle`, `class_name`) is immediately clear and requires no special knowledge to interpret.

3. **Unified cross-dataset format**: BOMNI and PIROPO annotations become identical in structure, enabling shared data loading code and consistent evaluation pipelines.

4. **Performance optimization**: Faster data loading during training (critical for large-scale experiments with multiple projection configurations).

Computational savings: For a dataset with N images evaluated over E epochs during training, our conversion eliminates $N \times E$ `arctan2()` calls. For PIROPO with 3,089 images evaluated 100 times (training and hyperparameter tuning), this saves approximately 309,000 trigonometric operations.

2.1.3 Frame Extraction and Filtering

Challenge: Original datasets contain thousands of frames per sequence, but Tamura only annotated sparse keyframes (∼10% of total frames).

Process:

1. Extract or organize frames from videos/downloads
2. Identify annotated frames by matching filenames with Tamura annotation files
3. **Delete all unannotated frames** (keeps only frames with verified ground truth)
4. Organize frames to match annotation directory hierarchy

Rationale: Unannotated frames may contain unlabeled pedestrians (false ground truth negatives) or be truly empty. Including them without verified annotations would compromise evaluation integrity. We conservatively use only frames with explicit ground truth labels.

2.1.4 Visualization Generation

Process:

1. Load each annotated frame and corresponding JSON annotations
2. Draw rotated bounding boxes on images using OpenCV
3. Save visualizations to dedicated output directory

Output: Visual representations of all annotations for human inspection

2.2 Manual Quality Review

User task:

1. Inspect all visualization images
2. Identify annotation errors:
   - Incorrect bbox position (misaligned with pedestrian)
   - Wrong size (too large/small, loose vs. tight fit inconsistency)
   - Incorrect rotation angle (not aligned with radial direction)
   - False positives (background objects mislabeled)
   - Missing detections (pedestrian visible but not annotated)
3. **Delete visualization images with incorrect annotations**

4. Keep only visualizations representing correct ground truth

Critical requirement: User deletes *only visualization images*, never annotation files directly. The automated cleanup step (Step 2) handles annotation removal.

2.3 Step 2: Automated Cleanup

Process:

1. Scan remaining visualization images (after user deletion)
2. Identify annotation files lacking corresponding visualizations (user deleted them)
3. **Delete orphaned annotation JSON files and frame images**
4. Regenerate final clean visualizations from verified annotations

Output: Production-ready dataset containing only manually verified correct annotations

Rationale: Manual verification filters annotation noise. Tamura annotations, like all manual labeling efforts, contain errors. Our BOMNI verification revealed a 25.5% error rate. Evaluating on uncorrected annotations yields misleading metrics influenced by annotation noise rather than true model performance.

3 BOMNI Dataset Processing

3.1 Dataset Overview

- **Source**: Bogazici Omni-directional and Multi-Camera Network Images
- **Environment**: Single indoor room, ceiling-mounted fisheye camera
- **Resolution**: 640×480 pixels
- **Fisheye center**: (320, 240)
- **Sequences**: scenario1 with 4 videos (top-0, top-1, top-2, top-3)
- **Tamura annotations**: 337 frames, 1,122 person instances (initial)

3.2 BOMNI-Specific Steps

3.2.1 Frame Extraction from Videos

- Input: 4 MP4 video files (top-0.mp4 through top-3.mp4)
- Extract all frames using OpenCV (cv2.VideoCapture)
- Naming: 4-digit serial numbers (0001.jpg, 0002.jpg, ...)
- Output: Thousands of frames per sequence

3.2.2 Cleanup and Conversion

- Delete unannotated frames (reduces from thousands to 337 annotated frames)
- Convert 337 Tamura XML files to standard JSON format
- Generate visualizations for all 337 frames

3.3 Manual Verification Results

- **Frames reviewed**: 337
- **Incorrect annotations identified**: 86
- **Error rate**: 25.5%
- **Error types**: Wrong position, inaccurate size, incorrect orientation, false positives
- **Final verified dataset**: 245 frames, 834 person instances (BOMNI-production)

3.4 Sequence Aggregation

Strategy: Combine all 4 sequences (top-0 through top-3) into unified BOMNI dataset
Justification:

- Per-sequence sample size insufficient (∼60 frames each) for reliable AP/precision/recall
- Identical camera position across sequences (same fisheye center, distortion model)
- Aggregation captures pedestrian diversity (appearance, pose, scale variation)
- Research goal: evaluate projection generalization, not sequence-specific tuning

3.5 BOMNI Limitations

1. **Limited scene variability**: Single room, constant lighting, uniform background
2. **Low resolution**: 640×480 (less detail than PIROPO's 800×600)
3. **Repetitive frames**: Near-duplicates from static pedestrians
4. **Small dataset**: 245 images (modest compared to COCO's 118k training images)
5. **Annotation noise**: 1–2% residual error expected even after manual correction

4 PIROPO Dataset Processing

4.1 Dataset Overview

- **Source**: People in Indoor ROoms with Perspective and Omnidirectional cameras
- **Rooms**: Room A (15×10m), Room B (8×5m)
- **Cameras**: 8 in Room A (3 omni + 5 perspective), 2 in Room B (1 omni + 1 perspective)

- **Resolution**: 800×600 pixels
- **Fisheye center**: (400, 300)
- **Tamura annotations**: 3,089 frames, 3,550 person instances (omni cameras only)

4.2 PIROPO-Specific Challenges

4.2.1 Massive Frame Archives

- Each camera folder: 2.8–4.9 GB (thousands of frames per sequence)
- Tamura annotated only sparse keyframes (∼10% of available frames)
- Unannotated frames drastically outnumber annotated ones

Solution: Extract *only* frames matching Tamura annotation timestamps (identified by ISO 8601 filenames, e.g., `2015-05-12T12-17-03.626Z.jpg`). Reduces dataset from ∼15 GB to ∼1 GB.

4.2.2 Hierarchical Folder Structure

- Original PIROPO: `Room A/omni_3A/omni_3A/omni3A_training/`
- Tamura annotations: `Room_A/omni_3A/omni_3A_training/`
- Differences: Spaces vs. underscores, duplicate camera folders, sequence naming

Solution: Reorganize extracted frames to match Tamura's flattened structure with standardized naming (`Room_A/{camera}/{sequence}/`).

4.3 Camera Selection

4.3.1 Excluded: Perspective Cameras

Cameras excluded: conv_4A through conv_8A (Room A), conv_2B (Room B)

Justification: Our research objective is fisheye pedestrian detection. Perspective cameras exhibit minimal radial distortion and do not require gnomonic projection transformations. Including perspective data would dilute evaluation metrics with out-of-scope samples.

4.3.2 Included: Omnidirectional Cameras Only

Cameras annotated by Tamura:

- Room_A: omni_1A, omni_2A, omni_3A
- Room_B: omni_1B

Total: 4 fisheye cameras, 3,089 frames (pre-verification)

4.4 Sequence Selection

4.4.1 Excluded: Unannotated Sequences

Sequences excluded:

- test4 through test12 (marked "SEQ" in PIROPO docs = sequences exist but no ground truth)
- training_seats (sitting pedestrians, not annotated by Tamura)

Justification: Quantitative evaluation requires ground truth. Sequences without Tamura annotations cannot contribute to precision, recall, AP, or IoU-based metrics.

4.4.2 Excluded: Empty Background Sequences

Sequences excluded: training_illum, training_bg, training_additional_bg

Content: Only background with illumination/appearance changes, no pedestrians

Justification: These sequences were provided for background modeling in classical motion-based detection (e.g., background subtraction). Our YOLO-based detector does not require background modeling. Frames without pedestrians contribute zero true positives and zero false negatives, providing no evaluation signal.

4.4.3 Included: Annotated Pedestrian Sequences

Sequences annotated by Tamura:

- **training**: Person walking exhaustively around room (3 people)
- **test2**: Up to 3 people simultaneously walking
- **test3**: 1 new person (not in training) walking

4.5 Camera Aggregation

Multi-camera structure:

- 4 physical cameras at different ceiling positions
- Same sequences recorded *simultaneously* from multiple viewpoints
- Same pedestrians captured from different angles and distances

Strategy: Combine all 3,089 annotations from all 4 cameras into unified PIROPO dataset

Justification:

1. **Statistical robustness**: Larger sample size (3,089 vs. ~750 per camera) for reliable metric estimates
2. **Viewpoint diversity**: Multiple angles stress-test projection configurations across varying radial distortion patterns
3. **Research alignment**: Goal is optimal configuration for *general* fisheye detection, not camera-specific tuning
4. **Consistency with BOMNI**: BOMNI aggregates sequences; PIROPO aggregates cameras. Both maximize data for robust evaluation.

Secondary analysis (optional): Per-camera performance breakdown as sanity check for camera-specific issues.

4.6 Manual Verification (Pending)

Following the same protocol as BOMNI:

1. Visualize all 3,089 frames
2. Manual review and deletion of incorrect annotations
3. Automated cleanup
4. Expected error rate: 10–25% (based on BOMNI experience)

5 Cross-Dataset Comparison

Characteristic	**BOMNI**	**PIROPO**
Cameras (fisheye)	1	4
Sequences per camera	4	3
Resolution	640×480	800×600
Fisheye center	(320, 240)	(400, 300)
Initial annotations	337	3,089
Person instances (initial)	1,122	3,550
Verified frames	245	TBD
Annotation error rate	25.5%	TBD
Environment	Single room	2 rooms
Scene complexity	Uniform background	Cluttered (Room B)
Processing pipeline	**Identical**	

Table 1: BOMNI vs. PIROPO dataset characteristics. Both follow the same two-step processing workflow with manual verification.

6 Methodological Consistency and Fair Evaluation

6.1 Why Consistency Matters

Applying identical processing to BOMNI and PIROPO ensures:

- Differences in projection configuration performance reflect *true dataset characteristics* (scene complexity, pedestrian scale, lighting) rather than processing artifacts
- Cross-dataset validation of optimal configurations (if Config A wins on both datasets, stronger evidence of generalization)
- Fair comparison: Same annotation quality standard (manual verification), same evaluation protocol

6.2 Conservative Data Inclusion

Our approach prioritizes **evaluation integrity over dataset size**:

- Exclude unannotated frames (avoid false negatives from missing labels)
- Exclude incorrect annotations (avoid rewarding wrong predictions)
- Exclude irrelevant data (perspective cameras, empty backgrounds)
- Result: Smaller but *cleaner* datasets with verified ground truth

6.3 Limitations and Mitigation

Small dataset sizes: BOMNI (245 images) and PIROPO (∼3,000 images) are modest compared to COCO (118k) or ImageNet (1.2M).

Mitigation strategies:

1. Focus on *relative* comparisons (projection configuration rankings) rather than absolute AP values
2. Cross-dataset validation (consistency across BOMNI and PIROPO increases confidence)
3. Conservative claims (acknowledge generalization limits beyond indoor fisheye scenarios)
4. Statistical significance testing (if sample size permits)

7 Summary

Both BOMNI and PIROPO datasets undergo rigorous two-step processing with manual quality verification:

1. **Step 1**: Automated extraction, filtering, conversion, and visualization
2. **Manual review**: Human inspection and deletion of incorrect annotations
3. **Step 2**: Automated cleanup and final dataset generation

Final production datasets contain only verified correct annotations:

- **BOMNI-production**: 245 frames, 834 person instances
- **PIROPO** (post-verification): TBD frames (∼2,300–2,700 expected)

This methodological rigor ensures evaluation metrics reflect true projection configuration performance rather than annotation noise or processing inconsistencies.

References

[1] Bastanlar, Y., and Akarun, L. *BOMNI: Bogazici Omni-directional and Multi-Camera Network Images Database*. Bogazici University, 2014. `https://www.cmpe.boun.edu.tr/pilab/pilabfiles/databases/bomni/`

[2] Gonzalez-Mora, J., et al. *PIROPO Database: People in Indoor ROoms with Perspective and Omnidirectional cameras*. PIROPO Project, 2014. `https://sites.google.com/site/piropodatabase/`

[3] Tamura, M., Horiguchi, S., and Murakami, T. *Omnidirectional Pedestrian Detection by Rotation Invariant Training*. 2019 IEEE Winter Conference on Applications of Computer Vision (WACV), pp. 1989–1998, 2019.